

INVENTION DISCLOSURE FORM



This form should be used by Researchers/Inventors of the **IIITDM**, and completed with a view to securing and obtaining patent protection for the same. This form is essentially designed to help the Researchers/Inventors to organize their thoughts about their invention in a structured manner. The Inventors are encouraged to explain their invention in an extremely simplistic manner and in complete terms so as to allow a person not familiar with the relevant technology to easily understand the disclosed invention. The disclosed invention would enable IP Attorneys in assessing its patentability and drafting a patent application on the same.

This invention disclosure form contains information that is proprietary and highly confidential. Unauthorized possession, viewing, or distribution of the form is strictly prohibited.

GUIDELINES TO FILL THE INVENTION DISCLOSURE FORM

Following are some guidelines that would help you in disclosing your invention in this Invention Disclosure Form (IDF):

- Please consider Attorneys as people who do not have any understanding of the technology in which your invention has been made, while disclosing the invention. Kindly disclose your invention to the fullest possible.
- Please do not limit yourself only to the specific method/process, formulation/ composition, product/device or prototype that you have invented, while disclosing your invention but also consider identifying all the possible modifications/alternatives of your invention. Kindly explain all the identified modifications/alternatives in detail as well.

Note:

• *Please remember your competitors could easily design around your invention and come up with alternatives/modification, which when not properly disclosed in the IDF may provide your competitors with competitive advantages over your invention.*

• *The best way you could think of the alternatives/modifications is to consider yourself as your own competitor!!*

- Inventors are encouraged to use ordinary and known technical terms to describe their invention. In case, some unusual terms are used then their appropriate definition must be provided.
- The invention must be explained with reference to the properly labeled drawings and flowcharts
- The IDF is divided into following sections: **Sections Purpose**

Sections	Purpose
Section- I	For disclosing details on the Applicant/Assignee and Inventor
Section-II	For disclosing details on the background of your invention
Section-III	For disclosing details of your invention

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SECTION-I: DETAILS OF THE ASSIGNEE/APPLICANT AND INVENTORS

Please provide below complete details of the Assignee/Applicant to which the invention is assigned by the inventors.

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Please provide details of all the inventors who have contributed to the invention. Feel free to add details of any additional inventors if required.

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1. Please write a brief background of the technology on which the invention has been made.

The invention relates to wearable navigation assistance systems using haptic feedback and wireless communication technologies such as Bluetooth Low Energy (BLE). It combines mobile navigation systems with embedded electronics to deliver directional cues without relying on visual or auditory interfaces.

2. Please write a brief background of the direct area/areas of application and use for which the invention has been made.

The invention is applicable in assistive navigation, wearable technology, and smart mobility systems. It is particularly useful for motorcyclists, cyclists, pedestrians, and visually impaired users who require non-distracting navigation guidance.

3. What are the existing problems of the technology that your invention proposes to solve? Have any previous attempts been made to solve these problems? Yes/No

If YES, how and by what means and what are their drawbacks/deficiencies?

Existing navigation systems rely heavily on visual displays or audio instructions, which can be unsafe or impractical in certain conditions such as driving or noisy environments. Previous attempts include audio navigation and simple vibration alerts; however:

- They lack precision in conveying direction and urgency
- They produce repetitive or redundant signals due to frequent GPS updates
- They are not optimized for wearable devices

4. How your invention proposes to overcome the above noted problems?

The proposed invention introduces an vibration-based navigation system that encodes directional and distance-based information into structured haptic signals transmitted via BLE. The system ensures minimal, non-redundant signaling using a state-based trigger mechanism.

5. Have you conducted any prior art search at your end on your invention? If YES, please provide us details of the prior art patents, published literature, catalogues, articles, etc. Please list down the patent or non-patent literature if you are aware of filed or published by academic and

non-academic groups working in the same or similar areas of application of your proposed invention.

Some prior systems include smartwatch navigation alerts and basic haptic feedback devices. However, these systems do not implement:

- Distance-tiered signal encoding
- One-time-trigger logic per navigation event
- Optimized signal abstraction for wearable interpretation

6. Who all could be the major research groups/competitors with IPR activities in India and abroad working specifically in the area of application of your invention? Please suggest some names. If any.

Companies such as Google (Wear OS navigation), Apple (Apple Watch navigation), and Garmin provide navigation assistance. However, their systems rely on generic haptic feedback and do not implement structured BLE-based directional encoding with state-aware triggering.

1. Please provide a short title of the invention in not more than 15 words. The title should be clear and succinct.

Wearable BLE-Based Directional Navigation System with Intelligent Haptic Signaling

2. Does your invention relate to an Apparatus/Device/System; Method/Process; Formulation/Composition; or a combination of the above? Please specify.

Combination of:

- Apparatus/Device/System
- Method/Process

3. What are the objectives of your invention?

- To provide distraction-free navigation using haptic feedback
- To reduce cognitive load during navigation
- To encode directional and distance information into minimal vibration signals
- To prevent redundant signaling using state tracking

4. Working of the invention:

The invention is a wearable navigation guidance system that captures navigation instructions from system notifications, processes them, and delivers real-time haptic feedback via a Bluetooth-enabled wearable device.

A. Navigation Data Acquisition Module

- The system uses a **notification listener service** to capture navigation instructions generated by external navigation applications
- Specifically, navigation text (e.g., "Turn right in 250 m") is intercepted from system notifications
- This approach eliminates dependency on proprietary APIs while enabling real-time instruction capture

B. Instruction Parsing Module

- The captured notification text is processed using a parsing engine
- The system extracts:

- Direction (left, right, U-turn, etc.)
 - Distance to next maneuver
- Keyword-based and rule-based parsing converts unstructured text into structured navigation instructions

C. Signal Encoding Module

- Parsed instructions are converted into compact signal codes
- The system uses a multi-layer proximity model:

Distance Range	Signal Layer	Example
> 500 m	Far	L1 / R1
100–250 m	Medium	L2 / R2
< 100 m	Near	L3 / R3

- Each signal encodes:
 - Direction (L/R)
 - Urgency level (1/2/3)

D. State-Based Triggering Module

- The system maintains internal state variables:
 - Current direction
 - Signal flags for each distance layer
- Each signal is triggered only once per navigation event:
 - Prevents repeated vibration due to frequent updates
- State resets when:
 - A new turn direction is detected
- This ensures controlled and non-redundant feedback

E. Bluetooth Communication Module (BLE)

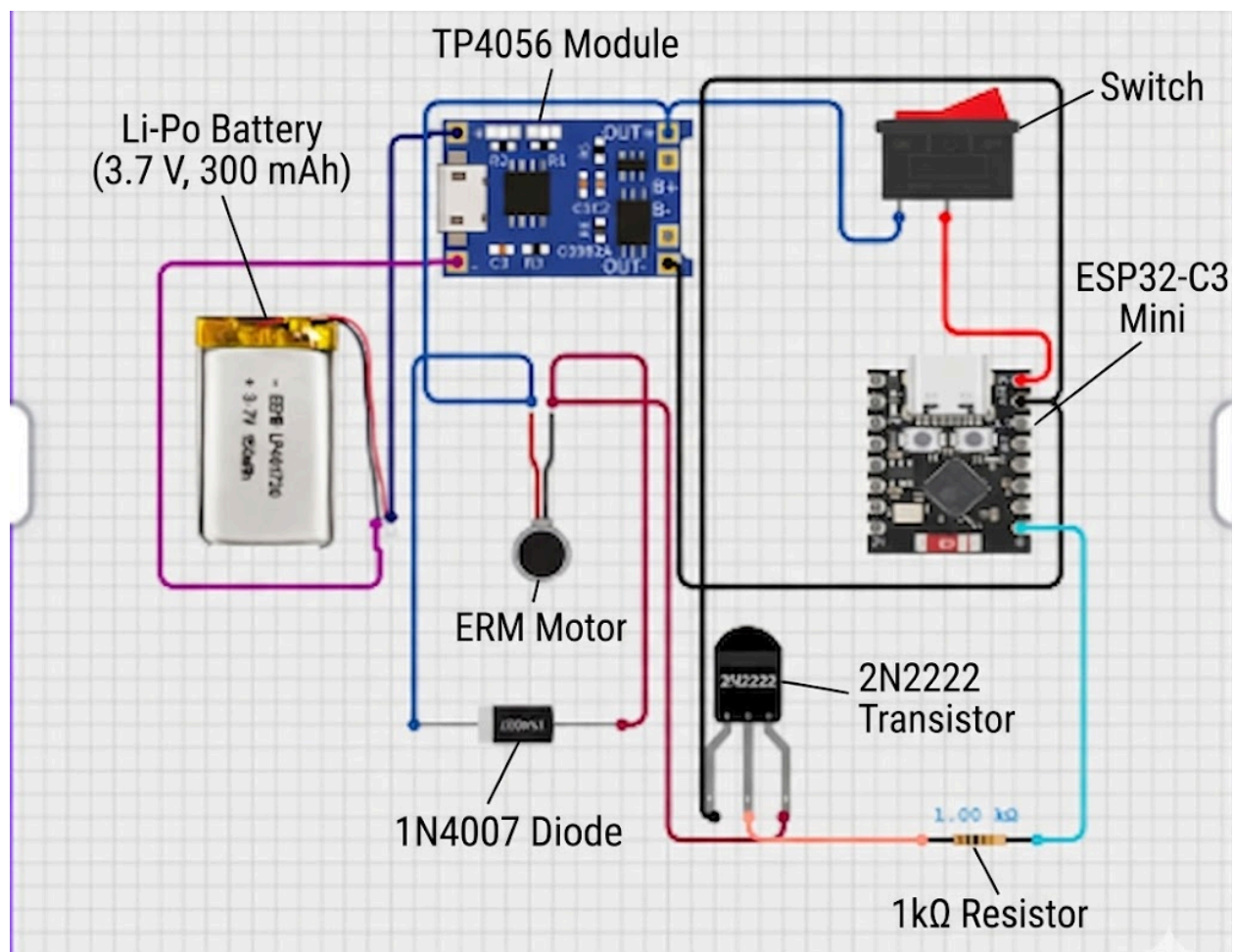
- Encoded signals are transmitted using Bluetooth Low Energy (BLE)
- Signals are sent as compact string commands (e.g., “L2”)
- Data is encoded as UTF-8 and written to a BLE characteristic
- Communication is optimized for low latency and low power consumption

F. Wearable Haptic Feedback Device

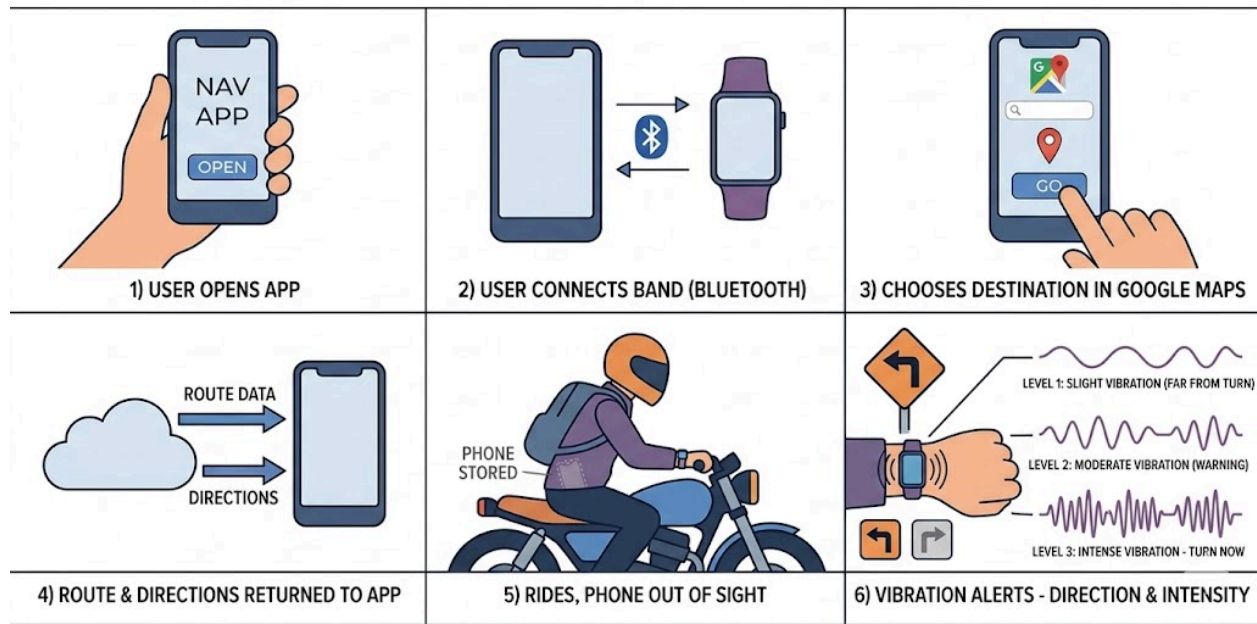
- A wearable device (ESP32-based) receives BLE signals
- The device decodes signals and activates vibration motors
- Different vibration patterns correspond to:
 - Left/right direction
 - Urgency level
- Example:
 - Left motor pulses → left turn
 - Right motor pulses → right turn
 - Both motors → alert (U-turn/arrival)

G. Operational Flow

1. Navigation app generates instruction notification
2. Notification listener captures text
3. Parser extracts direction and distance
4. Encoding module maps instruction to signal layer
5. State module checks if signal should be sent
6. Signal transmitted via BLE
7. Wearable device generates vibration pattern



SIMPLIFIED SMART WEARABLE NAVIGATION FLOW CHART



What are the novel features/steps of your invention?

- Extraction of navigation instructions directly from system notification streams without API integration
- Conversion of unstructured navigation text into structured directional signals
- Multi-layer distance-based haptic encoding system (L1, L2, L3)
- State-based one-time-trigger mechanism preventing redundant feedback
- Lightweight BLE communication using compact symbolic signals
- Integration of mobile software and wearable hardware for real-time navigation feedback

What are the advantages of your invention?

- Works with existing navigation apps without requiring API access
- Reduces dependency on proprietary services
- Provides distraction-free navigation via haptic feedback
- Minimizes signal redundancy and improves user experience
- Low power consumption due to efficient BLE communication
- Easily deployable on standard smartphones and microcontroller-based wearables

What is the primary business or product application of your invention? Are there other business or product application that might extend past your immediate focus?

Primary Application:

- The invention is primarily designed for **delivery riders** (e.g., food and parcel delivery personnel) who require continuous navigation while operating two-wheelers.
- It enables riders to receive navigation instructions through haptic feedback without needing to view mobile screens, thereby improving safety and efficiency during deliveries.

Examples include riders working with platforms such as Swiggy, Zomato, and Rapido.

Secondary Applications:

- Navigation assistance for motorcyclists and cyclists
- Pedestrian navigation in high-traffic environments
- Assistive technology for visually impaired users
- Silent navigation systems for military or tactical use
- Industrial worker guidance in hazardous environments

What could be the possible Modifications/Alternatives of your invention, if any?

- Use of direct navigation APIs instead of notification extraction
- Adaptive distance thresholds based on speed or environment
- Multi-point haptic feedback (different vibration zones)
- Integration with GPS/IMU sensors for independent navigation
- Advanced parsing using machine learning instead of keyword matching
- Multi-device ecosystem (helmet, gloves, etc.)

Has the invention been made and/or tested? Please provide details of any working example(s) that you would have.

The invention has been implemented as a working prototype consisting of:

- A Flutter-based mobile application utilizing notification listener services
- A BLE communication system for real-time signal transmission
- An ESP32-based wearable device for haptic feedback
- Successful testing of:
 - Notification-based instruction extraction
 - Signal encoding (L1, L2, L3, etc.)
 - One-time-trigger logic
 - Real-time vibration feedback





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